

[EECS 304] Control I

Sheet (10) Pole Placement

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Outline

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Problem Solution

Problem Solution is to get the output of the system in time domain if the input is known in the time domain.

Methods of solution:

1

$$\ell[u(t)] \rightarrow U(s)$$

$$Y(s) = U(s) * TF$$

$$\ell^{-1}[Y(s)] \rightarrow y(t)$$

2 We can get the weighting function of the system

$$h(t) = \ell^{-1}[TF]$$

$$y(t) = h(t) * u(t) = \int_0^t h(\tau)u(t - \tau)d\tau = \int_0^t u(\tau)h(t - \tau)d\tau$$

Problem Solution

Note:

- The previous two methods can be used when the system is SISO and we aren't interested in initial conditions.
- If we are interested in initial conditions or the system is MIMO, we will use a third method (SSR Method).

3 SSR Method:

$$\therefore \dot{\underline{x}} = A\underline{x} + B\underline{u}$$

$$\therefore \underline{y} = C\underline{x} + D\underline{u}$$

To solve this system, we will consider the scalar form of these equations:

$$\therefore \dot{x} = ax + bu$$

$$\therefore y = cx + du$$

Taking Laplace:

$$sX(s) - x(0) = aX(s) + bU(s)$$

$$\therefore (s - a)X(s) = bU(s) + x(0)$$

Problem Solution

$$\therefore X(s) = \frac{b}{s-a} U(s) + \frac{x(0)}{s-a}$$

Performing inverse Laplace:

$$\therefore x(t) = be^{at} * u(t) + x(0)e^{at}$$

$$\therefore x(t) = b \int_0^t e^{a(t-\tau)} u(\tau) d\tau + x(0)e^{at}$$

Returning to Matrix form:

$$\therefore \underline{x} = \int_0^t e^{A(t-\tau)} B u(\tau) d\tau + \underline{x}(0)e^{At}$$

After we get $\underline{x}$, we can get the output from the equation:

$$\underline{y} = C\underline{x} + D\underline{u}$$

Problem Solution

How to calculate e^{At} ?:

Let $Q(t)=e^{At}$ = State Transition Matrix

Using Taylor's expansion:

$$Q(t) = e^{At} = I + At + \frac{A^2}{2!}t^2 + + \frac{A^n}{n!}t^n +$$

$$\therefore Q(0) = I$$

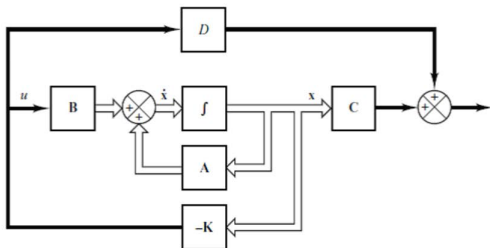
$$\therefore \frac{dQ(t)}{dt} = AQ(t)$$

$$\therefore sQ(s) - Q(0) = AQ(s)$$

$$\therefore (sI - A)Q(s) = Q(0) = I \rightarrow Q(s) = (sI - A)^{-1}$$

$$\therefore Q(t) = e^{At} = \ell^{-1}[sI - A]^{-1}$$

State Feedback



- State Feedback could be used to change the pole locations of a given system and hence, the unstable systems could be stabilized.

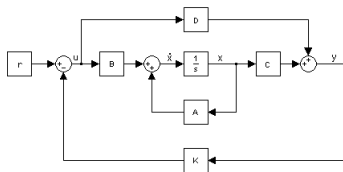
$$\therefore \dot{\underline{x}}(t) = A\underline{x}(t) + B u(t)$$

$$\therefore u = -k_1 x_1 - k_2 x_2 - k_3 x_3 = -\underline{k} \underline{x}$$

$$\therefore \dot{\underline{x}} = A\underline{x} + B\underline{u} = A\underline{x} + B(-\underline{k} \underline{x})$$

$$\therefore \dot{\underline{x}} = (A - B\underline{k}) \underline{x} = A_{new} \underline{x}$$

Output Feedback



$\therefore$ State Space Representation is

$$\dot{\underline{x}}(t) = A\underline{x}(t) + B\underline{u}(t)$$

$$\underline{y}(t) = C\underline{x}(t) + D\underline{u}(t)$$

For output feedback $\rightarrow u(t) = -ky(t)$

$$\therefore \dot{\underline{x}}(t) = A\underline{x}(t) + B(-ky(t)) = A\underline{x}(t) - Bky(t)$$

$$= A\underline{x}(t) - Bk[C\underline{x}(t) + D\underline{u}(t)]$$

$$\therefore \dot{\underline{x}}(t) = A\underline{x}(t) - BkC\underline{x}(t) - BkD\underline{u}(t)$$

$$= [A - BkC]\underline{x}(t) - BkD\underline{u}(t) = A_{new}\underline{x}(t) - BkD\underline{u}(t)$$

$\therefore$ New C/c equation $\rightarrow |sI - A_{new}| = 0$

Problem 1

Problem 1

A system is characterized by the state equation

$$\dot{\underline{x}} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

where $u(t)$ = unit step input. Compute the solution of the state vector if the initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$

Problem 1

Solution

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

$$e^{At} = \ell^{-1}\{[sI - A]^{-1}\}$$

$$[sI - A] = \begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}$$

$$[sI - A]^{-1} = \frac{1}{(s+1)^2} \begin{bmatrix} s+1 & 1 \\ 0 & s+1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} \\ 0 & \frac{1}{s+1} \end{bmatrix}$$

$$\ell^{-1}\{[sI - A]^{-1}\} = \begin{bmatrix} e^{-t} & te^{-t} \\ 0 & e^{-t} \end{bmatrix}$$

Problem 1

Solution

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

$$e^{At}x(0) = \begin{bmatrix} e^{-t} & te^{-t} \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} (1-t)e^{-t} \\ -e^{-t} \end{bmatrix}$$

$$\begin{aligned} \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau &= \int_0^t \begin{bmatrix} e^{-(t-\tau)} & (t-\tau)e^{-(t-\tau)} \\ 0 & e^{-(t-\tau)} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} d\tau \\ &= \int_0^t \begin{bmatrix} e^{-(t-\tau)} \\ 0 \end{bmatrix} d\tau = \int_0^t \begin{bmatrix} e^{-t}e^{\tau} \\ 0 \end{bmatrix} d\tau = \begin{bmatrix} e^{-t}e^{\tau} \\ 0 \end{bmatrix}_0^t = \begin{bmatrix} 1 - e^{-t} \\ 0 \end{bmatrix} \end{aligned}$$

$$\therefore x(t) = \begin{bmatrix} (1-t)e^{-t} \\ -e^{-t} \end{bmatrix} + \begin{bmatrix} 1 - e^{-t} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 - te^{-t} \\ -e^{-t} \end{bmatrix}$$

Problem 2

Problem 2

Compute the solution of the state vector for the system

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 2 \\ -3 & -5 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

given that $\underline{x}^T(0) = [3 \ 1]$ and $u(t)$ is a unit impulse.

Problem 2

Solution

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

$$e^{At} = \ell^{-1}\{[sI - A]^{-1}\}$$

$$[sI - A] = \begin{bmatrix} s & -2 \\ 3 & s+5 \end{bmatrix}$$

$$[sI - A]^{-1} = \frac{1}{s(s+5)+6} \begin{bmatrix} s+5 & 2 \\ -3 & s \end{bmatrix} = \begin{bmatrix} \frac{s+5}{s^2+5s+6} & \frac{2}{s^2+5s+6} \\ \frac{-3}{s^2+5s+6} & \frac{s}{s^2+5s+6} \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{3}{s+2} - \frac{2}{s+3} & \frac{2}{s+2} - \frac{2}{s+3} \\ \frac{-3}{s+2} + \frac{3}{s+3} & \frac{-2}{s+2} + \frac{3}{s+3} \end{bmatrix}$$

Problem 2

Solution

$$\therefore \ell^{-1}\{[sI - A]^{-1}\} = \begin{bmatrix} 3e^{-2t} - 2e^{-3t} & 2e^{-2t} - 2e^{-3t} \\ -3e^{-2t} + 3e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau = e^{At}x(0) + e^{At}B * u(t)$$

$$\begin{aligned} e^{At}x(0) &= e^{At} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3(3e^{-2t} - 2e^{-3t}) + 2e^{-2t} - 2e^{-3t} \\ 3(-3e^{-2t} + 3e^{-3t}) - 2e^{-2t} + 3e^{-3t} \end{bmatrix} \\ &= \begin{bmatrix} 11e^{-2t} - 8e^{-3t} \\ -11e^{-2t} + 12e^{-3t} \end{bmatrix} \end{aligned}$$

$$e^{At}B = e^{At} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2e^{-2t} - 2e^{-3t} \\ -2e^{-2t} + 3e^{-3t} \end{bmatrix}$$

$$e^{At}B * u(t) = \ell^{-1}\{\ell\{e^{At}B\}u(s)\}$$

Problem 2

Solution

$$= \ell^{-1} \left\{ \begin{bmatrix} \frac{2}{s+2} - \frac{2}{s+3} \\ \frac{-2}{s+2} + \frac{3}{s+3} \end{bmatrix} \right\}$$

$$\therefore e^{At} B * u(t) = e^{At} B$$

$$\therefore x(t) = e^{At} x(0) + e^{At} B * u(t)$$

$$\begin{aligned} &= \begin{bmatrix} 11e^{-2t} - 8e^{-3t} \\ -11e^{-2t} + 12e^{-3t} \end{bmatrix} + \begin{bmatrix} 2e^{-2t} - 2e^{-3t} \\ -2e^{-2t} + 3e^{-3t} \end{bmatrix} \\ &= \begin{bmatrix} 13e^{-2t} - 10e^{-3t} \\ -13e^{-2t} + 15e^{-3t} \end{bmatrix} \end{aligned}$$

Problem 3

Problem 3

Consider the system

$$\dot{\underline{x}} = A\underline{x} + B, \quad y = C\underline{x}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 0 \quad 0]$$

- Check the controllability of the system.
- Construct a control $u(t) = -k_1x_1(t) - k_2x_2(t) - k_3x_3(t)$, such that the characteristic equation of the system has roots at -2, -5 and -6. Determine the state feedback gain vector K and the control signal $u(t)$

Problem 3

Solution

Check controllability:

$$Q_c = [B|AB|A^2B]$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ -3 & -2 & 0 \\ 0 & -3 & -2 \end{bmatrix}$$

$$Q_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -2 \end{bmatrix} \rightarrow |Q_c| \neq 0$$

$\therefore \text{rank}(Q_c) \neq 0 \rightarrow \text{system is controllable}$

Problem 3

Solution

State Feedback:

$$\text{if } u = -k\underline{x}$$

$$\dot{\underline{x}} = A\underline{x} - Bk\underline{x} = (A - Bk)\underline{x} = A_{new}\underline{x}$$

$$A_{new} = A - Bk = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_1 - 3 & -k_2 - 2 & -k_3 \end{bmatrix}$$

$$\text{new } C/ch \text{ equation} = |sI - A_{new}|$$

$$= \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ k_1 + 3 & k_2 + 2 & s + k_3 \end{vmatrix}$$

$$= s[s(s + k_3) + k_2 + 2] + (k_1 + 3)$$

$$= s^3 + s^2(k_3) + s(k_2 + 2) + k_1 + 3 \rightarrow (1)$$

$$= (s + 2)(s + 5)(s + 6)$$

Problem 3

Solution

$$\begin{aligned} &= (s + 2)(s^2 + 11s + 30) \\ &= s^3 + 13s^2 + 52s + 60 \rightarrow (2) \end{aligned}$$

Comparing (1) & (2)

$$\therefore k_3 = 13, \quad k_2 = 50, \quad k_1 = 57$$

$$\therefore \underline{k} = [57 \quad 50 \quad 13] \rightarrow \text{gain vector}$$

$$\therefore \underline{u} = -57x_1(t) - 50x_2(t) - 13x_3(t) \rightarrow i/p \text{ control signal}$$

Problem 4

Problem 4

Consider the system

$$\dot{\underline{x}} = A\underline{x} + B, \quad y = C\underline{x}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 0 \quad 0]$$

- Check the controllability of the system.
- By using state feedback control $u(t) = -K\underline{x}(t)$, where K is a state feedback vector, it is desired to have the closed loop poles at:

$$\lambda_1 = -2 + j3.464, \quad \lambda_2 = -2 - j3.464, \quad \lambda_3 = -5$$

Determine the state feedback gain vector K and the control signal $u(t)$.

Problem 4

Solution

Check controllability:

$$Q_c = [B|AB|A^2B]$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 \\ -6 & -11 & -6 \\ 36 & 60 & 25 \end{bmatrix}$$

$$Q_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 25 \end{bmatrix} \rightarrow |Q_c| \neq 0$$

$\therefore \text{rank}(Q_c) \neq 0 \rightarrow \text{system is controllable}$

Problem 4

Solution

State Feedback:

$$\text{if } u = -k\underline{x}$$

$$\dot{\underline{x}} = A\underline{x} - Bk\underline{x} = (A - Bk)\underline{x} = A_{new}\underline{x}$$

$$A_{new} = A - Bk = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -k_1 - 6 & -k_2 - 11 & -k_3 - 6 \end{bmatrix}$$

$$\text{new } Cc_s \text{ equation} = |sI - A_{new}|$$

$$= \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ k_1 + 6 & k_2 + 6 & s + k_3 + 6 \end{vmatrix}$$

$$= s[s(s + k_3 + 6) + k_2 + 6] + k_1 + 6$$

$$= s^3 + s^2(k_3 + 6) + s(k_2 + 6) + k_1 + 6 \rightarrow (1)$$

$$= (s + 5)(s + 2 - j3.464)(s + 2 + j3.464)$$

Problem 4

Solution

$$\begin{aligned} &= (s + 5)(s^2 + 4s + 16) \\ &= s^3 + 9s^2 + 36s + 80 \rightarrow (2) \end{aligned}$$

Comparing (1) & (2)

$$\therefore k_3 = 3, \quad k_2 = 30, \quad k_1 = 74$$

$$\therefore \underline{k} = [74 \quad 30 \quad 3] \rightarrow \text{gain vector}$$

$$\therefore \underline{u} = -74x_1(t) - 30x_2(t) - 3x_3(t) \rightarrow i/p \text{ control signal}$$

Problem 5

Problem 5

Given:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 25 & 3 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 1 \quad 2]$$

- Can you use output feedback with feedback gain $K = 1$ to place a pole at $s = -4$? Verify your answer and find the poles of the overall system.
- If the A matrix was modified to become $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -23 & -25 & -7 \end{bmatrix}$, can you use output feedback with feedback gain $K = 1$ to place a pole at $s = -4$? Verify your answer and find the poles of the overall system.

Problem 5

Solution

a) For output feedback $\rightarrow u(t) = -ky(t) = -k(Cx(t))$

$$\therefore \dot{\underline{x}}(t) = A\underline{x}(t) + B[-kC\underline{x}(t)]$$

$$\therefore \dot{\underline{x}}(t) = (A - BkC)\underline{x}(t)$$

$\therefore$ C/ch equation $\rightarrow |sI - A + BkC| = 0$

$$\therefore \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 25 & 3 & -3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} [1 \quad 1 \quad 2] = 0$$

$$= \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ -25 & -3 & s+3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ -24 & -2 & s+5 \end{bmatrix} = 0$$

$$\therefore s(s(s+5) - 2) + 1(0 - 24) = 0$$

$$\therefore s^3 + 5s^2 - 2s - 24 = 0$$

$\therefore$ New poles $\rightarrow s = 2 \quad s = -3 \quad s = -4$

Problem 5

Solution

b)

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -23 & -25 & -7 \end{bmatrix}$$

$$\therefore |sI - A + BkC| = 0$$

$$\therefore \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 24 & 26 & s+9 \end{bmatrix} = 0$$

$$\therefore s(s(s+9) + 26) + 1(0 + 24) = 0$$

$$\therefore s^3 + 9s^2 + 26s + 24 = 0$$

$\therefore$ New poles $\rightarrow s = -2 \quad s = -3 \quad s = -4$

Thank you!

Questions?

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